

```
In [ ]: !git clone https://github.com/NTHung2034/PP-Final_Project.git
        %cd PP-Final_Project
```

```
Cloning into 'PP-Final_Project'...
remote: Enumerating objects: 1211, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (88/88), done.
remote: Total 1211 (delta 67), reused 72 (delta 35), pack-reused 1087 (from 1)
Receiving objects: 100% (1211/1211), 106.25 MiB | 39.02 MiB/s, done.
Resolving deltas: 100% (640/640), done.
/content/PP-Final_Project
```

```
In [ ]: !bash ./scripts/download_cifar10.sh
```

```
Downloading CIFAR-10 dataset...
Data directory: ./data
--2025-12-13 13:15:16-- https://www.cs.toronto.edu/~kriz/cifar-10-binary.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170052171 (162M) [application/x-gzip]
Saving to: 'cifar-10-binary.tar.gz'

cifar-10-binary.tar 100%[=====>] 162.17M  29.9MB/s   in 6.0s

2025-12-13 13:15:23 (26.8 MB/s) - 'cifar-10-binary.tar.gz' saved [170052171/170052171]

CIFAR-10 dataset downloaded and extracted to ./data
Files:
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_1.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_2.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_3.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_4.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_5.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 test_batch.bin
```

```
In [ ]: !mkdir build
        %cd build
```

```
/content/PP-Final_Project/build
```

```
In [ ]: %cd /content/PP-Final_Project/build
        !cmake .. -DENABLE_CUDA=ON
        # compile
        !make
```

```

/content/PP-Final_Project/build
-- The CXX compiler identification is GNU 11.4.0
-- The C compiler identification is GNU 11.4.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found CUDAToolkit: /usr/local/cuda/targets/x86_64-linux/include (found version "12.5.82")
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- CUDA Toolkit found - GPU support enabled
-- The CUDA compiler identification is NVIDIA 12.5.82 with host compiler GNU 11.4.0
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc - skipped
-- Detecting CUDA compile features
-- Detecting CUDA compile features - done
-- GPU Naive targets: train_gpu_naive, train_extract_naive, test_layers_gpu_naive, test_autoenc
oder_gpu_naive
-- GPU Optimized v1 targets: train_gpu_opt_v1, train_extract_opt_v1
-- GPU Optimized v2 targets: train_gpu_opt_v2, train_extract_opt_v2
-- Configuring done (4.2s)
-- Generating done (0.0s)
-- Build files have been written to: /content/PP-Final_Project/build
[ 1%] Building CXX object CMakeFiles/test_dataloader.dir/tests/test_dataloader.cpp.o
[ 2%] Building CXX object CMakeFiles/test_dataloader.dir/src/data/cifar10_loader.cpp.o
[ 3%] Linking CXX executable test_dataloader
[ 3%] Built target test_dataloader
[ 4%] Building CXX object CMakeFiles/test_layers_cpu.dir/tests/tests_cpu/test_layers.cpp.o
[ 5%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 6%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/relu_cpu.cpp.o
[ 7%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 8%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 9%] Linking CXX executable test_layers_cpu
[ 9%] Built target test_layers_cpu
[10%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/tests/tests_cpu/test_autoencod
er.cpp.o
[11%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/conv2d_cpu.cpp.o
[12%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/relu_cpu.cpp.o
[13%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/maxpool_cpu.cpp.o
[14%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/upsample_cpu.cpp.o
[15%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/models/autoencoder_cpu.cpp.o
[16%] Linking CXX executable test_autoencoder_cpu
[16%] Built target test_autoencoder_cpu
[17%] Building CXX object CMakeFiles/train_cpu.dir/src/train_cpu.cpp.o
[18%] Building CXX object CMakeFiles/train_cpu.dir/src/data/cifar10_loader.cpp.o
[19%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/conv2d_cpu.cpp.o
[20%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/relu_cpu.cpp.o
[21%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/maxpool_cpu.cpp.o
[22%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/upsample_cpu.cpp.o
[23%] Building CXX object CMakeFiles/train_cpu.dir/src/models/autoencoder_cpu.cpp.o
[24%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/logger.cpp.o
[25%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/image_utils.cpp.o
[26%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/memory_tracker.cpp.o
[27%] Linking CXX executable train_cpu

```

```
[ 27%] Built target train_cpu
[ 28%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/extract_features_cpu.cpp.o
[ 29%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/data/cifar10_loader.cpp.o
[ 30%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 31%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/relu_cpu.cpp.o
[ 32%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 33%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 34%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 35%] Linking CXX executable extract_features_cpu
[ 35%] Built target extract_features_cpu
[ 36%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/models_svm/train_svm_cpu.cpp.o
[ 37%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/data/cifar10_loader.cpp.o
[ 38%] Building CXX object CMakeFiles/train_svm_cpu.dir/external/svm.cpp.o
[ 39%] Linking CXX executable train_svm_cpu
[ 39%] Built target train_svm_cpu
[ 40%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/train_gpu_naive.cu.o
[ 41%] Building CXX object CMakeFiles/train_gpu_naive.dir/src/data/cifar10_loader.cpp.o
[ 42%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 43%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 44%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 45%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 46%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 47%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 48%] Linking CUDA device code CMakeFiles/train_gpu_naive.dir/cmake_device_link.o
[ 49%] Linking CXX executable train_gpu_naive
[ 49%] Built target train_gpu_naive
[ 50%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models_svm/train_extract_naive.cu.o
[ 51%] Building CXX object CMakeFiles/train_extract_naive.dir/src/data/cifar10_loader.cpp.o
[ 52%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 53%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 54%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 55%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 56%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 57%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 58%] Linking CUDA device code CMakeFiles/train_extract_naive.dir/cmake_device_link.o
[ 59%] Linking CXX executable train_extract_naive
[ 59%] Built target train_extract_naive
[ 60%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/train_gpu_opt_v1.cu.o
[ 61%] Building CXX object CMakeFiles/train_gpu_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 62%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 63%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 64%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 65%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 66%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
```

```

[ 67%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 68%] Linking CUDA device code CMakeFiles/train_gpu_opt_v1.dir/cmake_device_link.o
[ 69%] Linking CXX executable train_gpu_opt_v1
[ 69%] Built target train_gpu_opt_v1
[ 70%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models_svm/train_extract_opt_v1.cu.o
[ 71%] Building CXX object CMakeFiles/train_extract_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 72%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 73%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 74%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 75%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 76%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 77%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 78%] Linking CUDA device code CMakeFiles/train_extract_opt_v1.dir/cmake_device_link.o
[ 79%] Linking CXX executable train_extract_opt_v1
[ 79%] Built target train_extract_opt_v1
[ 80%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/train_gpu_opt_v2.cu.o
[ 81%] Building CXX object CMakeFiles/train_gpu_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 82%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 83%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 84%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 85%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 86%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 87%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 88%] Linking CUDA device code CMakeFiles/train_gpu_opt_v2.dir/cmake_device_link.o
[ 89%] Linking CXX executable train_gpu_opt_v2
[ 89%] Built target train_gpu_opt_v2
[ 90%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models_svm/train_extract_opt_v2.cu.o
[ 91%] Building CXX object CMakeFiles/train_extract_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 92%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 93%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 94%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 95%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 96%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 97%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 98%] Linking CUDA device code CMakeFiles/train_extract_opt_v2.dir/cmake_device_link.o
[100%] Linking CXX executable train_extract_opt_v2
[100%] Built target train_extract_opt_v2

```

```
In [ ]: !./train_gpu_opt_v2
```



# CIFAR-10 AUTOENCODER - GPU OPTIMIZED v2 (Fusion + Streams)

Using GPU: Tesla T4  
Compute Capability: 7.5  
Total Global Memory: 15095 MB

VRAM usage: 102 MB

Loading CIFAR-10 training data...  
Loaded data\_batch\_1.bin  
Loaded data\_batch\_2.bin  
Loaded data\_batch\_3.bin  
Loaded data\_batch\_4.bin  
Loaded data\_batch\_5.bin  
Training data loaded: 50000 images

## GPU OPTIMIZED v2 AUTOENCODER TRAINING (Kernel Fusion + Multi-Stream Pipeline)

### Configuration:

Batch size: 64  
Epochs: 20  
Learning rate: 0.001  
Training samples: 50000  
Save directory: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2

VRAM usage: 490 MB

Batches per epoch: 781

|                |                |               |                    |
|----------------|----------------|---------------|--------------------|
| Epoch [ 1/20]  | Loss: 0.060139 | Time: 173.98s | Avg: 173.98s/epoch |
| Epoch [ 2/20]  | Loss: 0.041295 | Time: 179.56s | Avg: 176.77s/epoch |
| Epoch [ 3/20]  | Loss: 0.035382 | Time: 179.93s | Avg: 177.82s/epoch |
| Epoch [ 4/20]  | Loss: 0.032324 | Time: 179.92s | Avg: 178.35s/epoch |
| Epoch [ 5/20]  | Loss: 0.030386 | Time: 179.78s | Avg: 178.63s/epoch |
| Epoch [ 6/20]  | Loss: 0.029086 | Time: 180.04s | Avg: 178.87s/epoch |
| Epoch [ 7/20]  | Loss: 0.028389 | Time: 179.71s | Avg: 178.99s/epoch |
| Epoch [ 8/20]  | Loss: 0.027530 | Time: 179.87s | Avg: 179.10s/epoch |
| Epoch [ 9/20]  | Loss: 0.027275 | Time: 179.65s | Avg: 179.16s/epoch |
| Epoch [ 10/20] | Loss: 0.027107 | Time: 179.78s | Avg: 179.22s/epoch |
| Epoch [ 11/20] | Loss: 0.026889 | Time: 179.83s | Avg: 179.28s/epoch |
| Epoch [ 12/20] | Loss: 0.026607 | Time: 179.74s | Avg: 179.31s/epoch |
| Epoch [ 13/20] | Loss: 0.026265 | Time: 179.74s | Avg: 179.35s/epoch |
| Epoch [ 14/20] | Loss: 0.026325 | Time: 179.62s | Avg: 179.37s/epoch |
| Epoch [ 15/20] | Loss: 0.026022 | Time: 179.60s | Avg: 179.38s/epoch |
| Epoch [ 16/20] | Loss: 0.025674 | Time: 179.81s | Avg: 179.41s/epoch |
| Epoch [ 17/20] | Loss: 0.025882 | Time: 179.48s | Avg: 179.41s/epoch |
| Epoch [ 18/20] | Loss: 0.025428 | Time: 179.86s | Avg: 179.44s/epoch |
| Epoch [ 19/20] | Loss: 0.025363 | Time: 179.56s | Avg: 179.44s/epoch |
| Epoch [ 20/20] | Loss: 0.025113 | Time: 179.87s | Avg: 179.47s/epoch |

### TRAINING COMPLETED

Total training time: 3589.32s  
Average time per epoch: 179.47s  
Initial loss: 0.060139

Final loss: 0.025113  
Loss reduction: 58.242218%

=====

Training summary saved to: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/training\_summary.txt  
Final VRAM usage: 490 MB

TRAINING COMPLETED!

In [ ]: !./train\_extract\_opt\_v2

## CIFAR-10 FEATURE EXTRACTION - GPU OPT V2 (PHASE 4)

=== Step 1: Loading Dataset ===

Loading CIFAR-10 training data...

Loaded data\_batch\_1.bin

Loaded data\_batch\_2.bin

Loaded data\_batch\_3.bin

Loaded data\_batch\_4.bin

Loaded data\_batch\_5.bin

Training data loaded: 50000 images

Loading CIFAR-10 test data...

Test data loaded: 10000 images

Training images: 50000

Test images: 10000

=== Step 2: Loading Encoder Weights ===

Encoder weights loaded successfully

=== Step 3: Extracting Features ===

Train batch [50/500]

Train batch [100/500]

Train batch [150/500]

Train batch [200/500]

Train batch [250/500]

Train batch [300/500]

Train batch [350/500]

Train batch [400/500]

Train batch [450/500]

Train batch [500/500]

Test batch [20/100]

Test batch [40/100]

Test batch [60/100]

Test batch [80/100]

Test batch [100/100]

Feature extraction completed: 26.39s

=== Step 4: Saving Features to .bin Files ===

Saved train features: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/train\_features.bin

Shape: [50000, 8192]

Saved train labels: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/train\_labels.bin

Shape: [50000]

Saved test features: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/test\_features.bin

Shape: [10000, 8192]

Saved test labels: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/test\_labels.bin

Shape: [10000]

Saved metadata: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/metadata.txt

=== Summary ===

Feature extraction time: 26.39s

All features saved to: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features

FEATURE EXTRACTION COMPLETED!

```
In [ ]: %cd /content/PP-Final_Project/  
!python /content/PP-Final_Project/src/models_svm/cuml_svm_train.py opt_v2
```

/content/PP-Final\_Project

```
=====
CIFAR-10 cudaML SVM Training with RBF Kernel [OPT_V2]
=====
```

Configuration:

Implementation: OPT\_V2  
Data directory: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features  
Kernel: rbf  
C: 10.0  
Gamma: auto

=== Step 1: Loading Data ===

Loaded features shape: (50000, 8192)  
Loaded labels shape: (50000,)  
Loaded features shape: (10000, 8192)  
Loaded labels shape: (10000,)  
Train samples: 50000  
Test samples: 10000

Normalizing features...

Features normalized (mean=0, std=1)

=== Step 2: Training SVM with cudaML ===

Initializing SVM classifier...

Training started...

[2025-12-13 14:18:53.080] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:53.949] [CUMML] [debug] SMO solver finished after 44 outer iterations, total inner 23090 iterations, and diff 0.000995  
[2025-12-13 14:18:54.547] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:55.271] [CUMML] [debug] SMO solver finished after 55 outer iterations, total inner 31455 iterations, and diff 0.000978  
[2025-12-13 14:18:55.771] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:56.371] [CUMML] [debug] SMO solver finished after 43 outer iterations, total inner 23456 iterations, and diff 0.000996  
[2025-12-13 14:18:56.869] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:57.465] [CUMML] [debug] SMO solver finished after 43 outer iterations, total inner 25220 iterations, and diff 0.000994  
[2025-12-13 14:18:57.965] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:58.472] [CUMML] [debug] SMO solver finished after 37 outer iterations, total inner 19483 iterations, and diff 0.000984  
[2025-12-13 14:18:58.975] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:18:59.386] [CUMML] [debug] SMO solver finished after 30 outer iterations, total inner 15323 iterations, and diff 0.000967  
[2025-12-13 14:18:59.899] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:00.451] [CUMML] [debug] SMO solver finished after 40 outer iterations, total inner 21497 iterations, and diff 0.000984  
[2025-12-13 14:19:00.947] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:01.805] [CUMML] [debug] SMO solver finished after 65 outer iterations, total inner 38574 iterations, and diff 0.000989  
[2025-12-13 14:19:02.303] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:02.985] [CUMML] [debug] SMO solver finished after 51 outer iterations, total inner 26205 iterations, and diff 0.000978  
[2025-12-13 14:19:03.483] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:03.969] [CUMML] [debug] SMO solver finished after 33 outer iterations, total inner 15921 iterations, and diff 0.000987  
[2025-12-13 14:19:04.465] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:05.044] [CUMML] [debug] SMO solver finished after 39 outer iterations, total inner 18511 iterations, and diff 0.000995  
[2025-12-13 14:19:05.533] [CUMML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:05.980] [CUMML] [debug] SMO solver finished after 31 outer iterations, total inner



nner 15366 iterations, and diff 0.000982  
[2025-12-13 14:19:06.476] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:06.977] [CUML] [debug] SMO solver finished after 33 outer iterations, total i  
nner 15846 iterations, and diff 0.000997  
[2025-12-13 14:19:07.478] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:07.950] [CUML] [debug] SMO solver finished after 32 outer iterations, total i  
nner 15605 iterations, and diff 0.001000  
[2025-12-13 14:19:08.448] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:08.999] [CUML] [debug] SMO solver finished after 38 outer iterations, total i  
nner 18316 iterations, and diff 0.000993  
[2025-12-13 14:19:09.506] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:10.188] [CUML] [debug] SMO solver finished after 48 outer iterations, total i  
nner 24979 iterations, and diff 0.000998  
[2025-12-13 14:19:10.695] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:11.764] [CUML] [debug] SMO solver finished after 78 outer iterations, total i  
nner 39301 iterations, and diff 0.000983  
[2025-12-13 14:19:12.275] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:13.337] [CUML] [debug] SMO solver finished after 80 outer iterations, total i  
nner 46918 iterations, and diff 0.000996  
[2025-12-13 14:19:13.839] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:14.953] [CUML] [debug] SMO solver finished after 85 outer iterations, total i  
nner 53552 iterations, and diff 0.000988  
[2025-12-13 14:19:15.462] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:16.479] [CUML] [debug] SMO solver finished after 75 outer iterations, total i  
nner 43980 iterations, and diff 0.000982  
[2025-12-13 14:19:16.974] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:17.792] [CUML] [debug] SMO solver finished after 63 outer iterations, total i  
nner 37720 iterations, and diff 0.000959  
[2025-12-13 14:19:18.288] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:19.123] [CUML] [debug] SMO solver finished after 63 outer iterations, total i  
nner 35417 iterations, and diff 0.000942  
[2025-12-13 14:19:19.627] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:20.155] [CUML] [debug] SMO solver finished after 38 outer iterations, total i  
nner 20391 iterations, and diff 0.000998  
[2025-12-13 14:19:20.648] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:21.167] [CUML] [debug] SMO solver finished after 36 outer iterations, total i  
nner 18054 iterations, and diff 0.000997  
[2025-12-13 14:19:21.679] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:22.662] [CUML] [debug] SMO solver finished after 74 outer iterations, total i  
nner 43212 iterations, and diff 0.000986  
[2025-12-13 14:19:23.158] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:24.534] [CUML] [debug] SMO solver finished after 104 outer iterations, total  
inner 58320 iterations, and diff 0.000995  
[2025-12-13 14:19:25.034] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:26.006] [CUML] [debug] SMO solver finished after 71 outer iterations, total i  
nner 40180 iterations, and diff 0.000984  
[2025-12-13 14:19:26.515] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:27.448] [CUML] [debug] SMO solver finished after 66 outer iterations, total i  
nner 36490 iterations, and diff 0.000986  
[2025-12-13 14:19:27.950] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:28.476] [CUML] [debug] SMO solver finished after 37 outer iterations, total i  
nner 18539 iterations, and diff 0.000984  
[2025-12-13 14:19:28.971] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:29.595] [CUML] [debug] SMO solver finished after 42 outer iterations, total i  
nner 21691 iterations, and diff 0.000987  
[2025-12-13 14:19:30.094] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:31.005] [CUML] [debug] SMO solver finished after 67 outer iterations, total i  
nner 40094 iterations, and diff 0.000980  
[2025-12-13 14:19:31.511] [CUML] [debug] Creating working set with 1024 elements  
[2025-12-13 14:19:32.415] [CUML] [debug] SMO solver finished after 65 outer iterations, total i  
nner 40102 iterations, and diff 0.000994  
[2025-12-13 14:19:32.921] [CUML] [debug] Creating working set with 1024 elements

```
[2025-12-13 14:19:33.882] [CUML] [debug] SMO solver finished after 71 outer iterations, total i
nner 41207 iterations, and diff 0.000990
[2025-12-13 14:19:34.381] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:34.861] [CUML] [debug] SMO solver finished after 33 outer iterations, total i
nner 17913 iterations, and diff 0.000994
[2025-12-13 14:19:35.353] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:35.854] [CUML] [debug] SMO solver finished after 36 outer iterations, total i
nner 17700 iterations, and diff 0.000990
[2025-12-13 14:19:36.363] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:37.151] [CUML] [debug] SMO solver finished after 56 outer iterations, total i
nner 32560 iterations, and diff 0.000998
[2025-12-13 14:19:37.657] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:38.607] [CUML] [debug] SMO solver finished after 69 outer iterations, total i
nner 37881 iterations, and diff 0.000993
[2025-12-13 14:19:39.108] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:39.552] [CUML] [debug] SMO solver finished after 31 outer iterations, total i
nner 15443 iterations, and diff 0.000981
[2025-12-13 14:19:40.049] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:40.595] [CUML] [debug] SMO solver finished after 36 outer iterations, total i
nner 17811 iterations, and diff 0.000999
[2025-12-13 14:19:41.104] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:41.740] [CUML] [debug] SMO solver finished after 45 outer iterations, total i
nner 24703 iterations, and diff 0.000983
[2025-12-13 14:19:42.236] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:42.619] [CUML] [debug] SMO solver finished after 28 outer iterations, total i
nner 12826 iterations, and diff 0.000997
[2025-12-13 14:19:43.115] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:43.627] [CUML] [debug] SMO solver finished after 35 outer iterations, total i
nner 17102 iterations, and diff 0.000975
[2025-12-13 14:19:44.138] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:44.597] [CUML] [debug] SMO solver finished after 31 outer iterations, total i
nner 15615 iterations, and diff 0.000974
[2025-12-13 14:19:45.098] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:45.724] [CUML] [debug] SMO solver finished after 42 outer iterations, total i
nner 21654 iterations, and diff 0.000989
[2025-12-13 14:19:46.222] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 14:19:46.886] [CUML] [debug] SMO solver finished after 45 outer iterations, total i
nner 23856 iterations, and diff 0.000997
Training completed in 62.82s
```

=== Step 3: Evaluation ===

Evaluating on training set...

Processed 10000/50000 samples

Processed 20000/50000 samples

Processed 30000/50000 samples

Processed 40000/50000 samples

Processed 50000/50000 samples

Evaluating on test set...

Training Accuracy: 98.66%

Test Accuracy: 66.70%

Confusion matrix saved to: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_featur  
es/confusion\_matrix.png

=== Per-Class Accuracy ===

airplane: 71.90%

automobile: 76.30%

bird: 52.60%

cat: 52.80%

deer: 62.90%

dog: 57.60%

frog: 73.60%

horse: 70.30%  
ship: 77.00%  
truck: 72.00%

=== Classification Report ===

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| airplane     | 0.7277    | 0.7190 | 0.7233   | 1000    |
| automobile   | 0.7638    | 0.7630 | 0.7634   | 1000    |
| bird         | 0.5984    | 0.5260 | 0.5599   | 1000    |
| cat          | 0.4248    | 0.5280 | 0.4708   | 1000    |
| deer         | 0.6485    | 0.6290 | 0.6386   | 1000    |
| dog          | 0.5957    | 0.5760 | 0.5857   | 1000    |
| frog         | 0.7434    | 0.7360 | 0.7397   | 1000    |
| horse        | 0.7369    | 0.7030 | 0.7195   | 1000    |
| ship         | 0.7708    | 0.7700 | 0.7704   | 1000    |
| truck        | 0.7122    | 0.7200 | 0.7161   | 1000    |
| accuracy     |           |        | 0.6670   | 10000   |
| macro avg    | 0.6722    | 0.6670 | 0.6687   | 10000   |
| weighted avg | 0.6722    | 0.6670 | 0.6687   | 10000   |

=== Step 4: Saving Model ===

Model saved to: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/cuml\_svm\_model.pkl

=====

TRAINING SUMMARY

=====

Configuration:

Implementation: OPT\_V2  
Kernel: rbf  
C: 10.0  
Gamma: auto

Training time: 62.82s  
Training accuracy: 98.66%  
Test accuracy: 66.70%

=====

=== Per-Class Accuracy ===

airplane: 71.90%  
automobile: 76.30%  
bird: 52.60%  
cat: 52.80%  
deer: 62.90%  
dog: 57.60%  
frog: 73.60%  
horse: 70.30%  
ship: 77.00%  
truck: 72.00%

=== Classification Report ===

|            | precision | recall | f1-score | support |
|------------|-----------|--------|----------|---------|
| airplane   | 0.7277    | 0.7190 | 0.7233   | 1000    |
| automobile | 0.7638    | 0.7630 | 0.7634   | 1000    |
| bird       | 0.5984    | 0.5260 | 0.5599   | 1000    |

|              |        |        |        |       |
|--------------|--------|--------|--------|-------|
| cat          | 0.4248 | 0.5280 | 0.4708 | 1000  |
| deer         | 0.6485 | 0.6290 | 0.6386 | 1000  |
| dog          | 0.5957 | 0.5760 | 0.5857 | 1000  |
| frog         | 0.7434 | 0.7360 | 0.7397 | 1000  |
| horse        | 0.7369 | 0.7030 | 0.7195 | 1000  |
| ship         | 0.7708 | 0.7700 | 0.7704 | 1000  |
| truck        | 0.7122 | 0.7200 | 0.7161 | 1000  |
| accuracy     |        |        | 0.6670 | 10000 |
| macro avg    | 0.6722 | 0.6670 | 0.6687 | 10000 |
| weighted avg | 0.6722 | 0.6670 | 0.6687 | 10000 |

Training summary saved to: /content/PP-Final\_Project/models/saved\_weights\_gpu\_opt\_v2/svm\_features/training\_summary.txt

```
In [ ]: !zip -r models_v2.zip /content/PP-Final_Project/models
```

```
adding: content/PP-Final_Project/models/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights/epoch_metrics_cpu.txt (deflated 3%)
adding: content/PP-Final_Project/models/saved_weights/svm_results.txt (deflated 47%)
adding: content/PP-Final_Project/models/saved_weights/test_predictions.txt (deflated 65%)
adding: content/PP-Final_Project/models/saved_weights/cpu_final.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_weights.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights/train_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/training_summary.txt (deflated 41%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv3_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv3_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv5_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv2_w.bin (deflated 9%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv2_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv1_b.bin (deflated 1%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv1_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/training_summary.txt (deflated 66%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/test_labels.bin (deflated 49%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/test_features.bin (deflated 41%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/metadata.txt (deflated 24%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/confusion_matrix.png (deflated 16%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/train_labels.bin (deflated 51%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/cuml_svm_model.pkl (deflated 40%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/svm_features/train_features.bin (deflated 41%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv5_b.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv4_w.bin (deflated 8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/conv4_b.bin (deflated 1%)
adding: content/PP-Final_Project/models/saved_weights_gpu_optimized/ (stored 0%)
```

```
In [ ]: from google.colab import files
files.download("models_v2.zip")
```